

MEASURING PERSUASTION WITHOUT MEASURING A PRIOR BELIEF:
A NEW APPLICATION OF PLANNED MISSING DATA TECHNIQUES

BY

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DISSERTATION

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF DOCTOR OF PHILOSOPHY
IN THE DEPARTMENT OF PSYCHOLOGY
AT FORDHAM UNIVERSITY

NEW YORK

November 13, 2023

To Elena, for being such a solid foundation even when the architecture on top of it got a bit unstable; and to Amy, for never completely losing hope I'd become a bona fide academic after all.

Acknowledgements

This dissertation would not have been possible without the support of the entire faculty of the Psychometrics and Quantitative Psychology program at Fordham University. My mentor and advisor, Dr. David Budescu, for his insightful feedback that he was eager to share when requested, even when those requests were large and frequent; as well as his patience in waiting for those requests to materialize when they were not frequent. Dr. Heining Cham, for having been the instructor for more hours of my statistical training than anyone else, and for giving me permission to use the computers in his lab to run time-consuming simulations. And Dr. Leah Feuerstahler, for the numerous occasions on which she provided invaluable wisdom in both my professional and personal lives, and for patiently and collaboratively ensuring I made the mathematical notation in this dissertation make sense.

I would also like to thank the entire SAGE team of the Hybrid Forecasting Competition, who gave me a tremendous leg up in getting my research career off the ground.

This Dissertation was funded by The National Science Foundation Doctoral Dissertation Research Improvement Grant in Decision, Risk and Management Sciences, Award #2242100; as well as the Fordham University Anne Anastasi chair.

This Dissertation is the product of a labor union: Fordham Graduate Student Workers, Communication Workers of America, Local 1104.

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PREVIEW

Chapter 1: Introduction

Thesis

The *observer effect* posits that the process of measuring a phenomenon fundamentally alters it (Baclawski, 2018). In psychological studies, this is also sometimes referred to as *measurement reactivity* (Arslan et al., 2021; Harel et al., 2015). One example of this effect occurs in studies of advice utilization. A classic advice utilization paradigm is known as a *judge advisor system* (JAS; Bailey et al., 2023; Himmelstein, 2022; Himmelstein & Budescu, 2022, 2023; Logg et al., 2019; Önköl et al., 2009; Schultze et al., 2015; Snizek & Buckley, 1995). In a JAS, a judge is asked to express a belief or judgment; for example, to estimate how many calories are in a particular food item, predict the probability that a football team will win the Super Bowl, or forecast the exchange rate between the U.S. Dollar and the Euro a year into the future. The judge is then offered an alternative opinion, courtesy of one (or more), advisor(s), which could be any source of independent information, and allowed to revise their original judgment. The influence of the advice can be measured from the observed change between the original and revised judgments.

However, there is a critical design feature of the JAS paradigm that limits its generalizability outside of laboratory settings: it requires that a judge express their prior belief directly and explicitly before receiving advice. There are many scenarios in which people don't make their beliefs explicit before evaluating new information. For example, if someone is trying to decide whether they want to watch a particular movie, they might check its score on Rotten Tomatoes, but they won't necessarily have to estimate its exact score before doing so. Or perhaps a graduate school admissions office uses an algorithm to rate applicants. Individual admissions officers typically won't try to estimate the precise rank of each candidate prior to

checking the results of their system. In fact, there is evidence judges' behavior is dependent on whether these prior beliefs were made explicit or not (Koehler & Beaugregard, 2006; Rader et al., 2015; Snizek & Buckley, 1995; Yaniv & Chosen-Hillel, 2012). Given that, there is no way to know if even robust patterns found in JAS research can tell us about behavior in these common real-world examples, because these judges never recorded what they believed in advance.

How can we measure change in something if we did not measure its state, or value, prior to the change? This question seems like an impossible counterfactual. My research intends to simultaneously address two problems. First, it will subvert measurement reactivity and describe the psychological process of advice utilization in the absence of explicit prior belief. Second, it will demonstrate the use of modern quantitative methods to measure change in the absence of direct measurement of the baseline from which the change occurred.

Literature Review

Measuring Belief Change

Advice utilization in JAS studies is commonly measured with a metric called *weight of advice* (WOA; Bailey et al., 2023; Bonaccio & Dalal, 2006; Harvey & Fischer, 1997; Himmelstein, 2022; Himmelstein & Budescu, 2023; Logg et al., 2019; Önköl et al., 2009; Schultze et al., 2015), a shift proportion representing the change in a judge's belief. Let J_1 be a judge's initial judgment, J_a the advice, and J_2 the judge's revised judgment¹. WOA can be calculated

$$\text{WOA} = \frac{J_2 - J_1}{J_a - J_1} \quad (1)$$

¹ Henceforth, the terms J_1 , initial judgment, and prior judgment will be used interchangeably; the terms J_2 , revised judgment, and posterior judgment will be used interchangeably; and the terms J_a and advice will be used interchangeably.

The logic of WOA is that J_2 is a weighted average of J_1 and J_a . Indeed, rearranging terms, this formula implies

$$J_2 = WOA(J_a) + (1 - WOA)(J_1) \quad (2)$$

In this formulation, WOA is a weighting parameter that determines the extent to which J_2 reflects J_a as opposed to J_1 . Normatively, it can be shown that if both the judge's initial belief and the advice are good faith estimates of both agents' true beliefs, to optimize J_2 for accuracy, WOA must be $\in [0, 1]$ (Larrick & Soll, 2006; Soll & Larrick, 2009). Even if one thinks the advice is useless, it should not be *dissuasive*—the judge's belief should simply remain what it was before seeing the advice. Likewise, if one's initial belief was a random guess, simply adopting the advice is strictly superior to a revised belief that discounts for the initial random guess. In many studies, a small proportion of WOA values lie outside of this normative interval, which are often either Winsorized or discarded (Bonaccio & Dalal, 2006; Himmelstein & Budescu, 2023; Logg et al., 2019; Schultze et al., 2015). Alternatively, in some cases the absolute differences are used, such that we can define absolute WOA (aWOA)

$$aWOA = \frac{|J_2 - J_1|}{|J_a - J_1|} \quad (3)$$

Meta-Analysis

A recent meta-analysis (Bailey et al., 2023) provides a thorough systematic review of high-level insights generated by JAS research across 129 unique datasets. Overall, the pooled mean WOA across all studies surveyed was 0.39, indicating people favored their prior beliefs about 1.5 times more heavily than advice. However, the degree of heterogeneity across studies was high, ranging from around 0.10 to around 0.75. Among meaningful determinants of advice utilization, they found the strongest predictor to be the perceived quality of the advice or advisor,

with higher quality advisors leading to greater advice utilization. The objectivity of the estimation target also influenced advice utilization. Objective tasks, which had a known ground truth, were separated from subjective or uncertain tasks, which either lacked a ground truth entirely, or did not have one at the time of elicitation (such as forecasts). Results indicated advice utilization tended to be higher for subjective or uncertain tasks than objective tasks. However, demographic variables about the judges themselves, such as age and gender, had little to no association with how heavily advice was utilized.

Egocentric Bias

A consistent effect found throughout the literature is known as *egocentric trimming* or *egocentric discounting*, where people weigh their own prior beliefs more heavily than advice. Prior to the meta-analysis by Bailey et al. (2023), less systematic reviews typically found mean *WOA* to be in the range of .2 to .3 (Harvey & Fischer, 1997; Soll & Larrick, 1999, 2009; Yaniv & Kleinberger, 2000), meaning that, on average, people weigh their prior beliefs two to four times as heavily as advice. And although Bailey et al. (2023) found the pooled mean *WOA* to be slightly higher, they still found that it was weighted more towards the self rather than advice.

Another important phenomenon identified by JAS research is that the (dis)similarity between J_1 and J_a has a meaningful impact on how heavily advice tends to be weighted: people tend to more heavily discount advice that is highly dissimilar from their initial beliefs (Allahverdyan & Galstyan, 2014; Himmelstein & Budescu, 2023; Schultze et al., 2015; Yaniv, 1997, 2004; Yaniv & Milyavsky, 2007). More recent research has also resurrected a different classic finding: people also discount advice that is *too* similar to their own beliefs (Himmelstein & Budescu, 2023; Schultze et al., 2015; Whittaker, 1963). Presumably, in such cases, when advice is only slightly different from one's prior belief, it is treated as a confirmation, rather than

new information to be integrated into a posterior belief. The result is an apparent curvilinear pattern, such that people's belief revisions are largest for cases where J_a is moderately different from J_1 , and smaller when they are either highly similar or highly different.

Probability Judgment

A special class of JAS literature involves probability judgment (Himmelstein & Budescu, 2023; Logg et al., 2019; Soll & Larrick, 2009). This could refer to cases where judges try to guess factual proportions about known quantities (e.g., what is the proportion of the population of a given country that smokes cigarettes; Soll & Larrick, 2009), or forecasts about the relative likelihoods of future events (e.g., what is the probability a proposed law will pass by a given date; Himmelstein & Budescu, 2023; Logg et al., 2019). Many cases focus on binary events, where judges predict the likelihood that a given event will or will not occur (Logg et al., 2019; Soll & Larrick, 2009). Because $P(\text{event occurs}) = 1 - P(\text{event does not occur})$, *WOA* easily generalizes to such binary probability judgments.

Himmelstein and Budescu (2023) explored a generalization of JAS research for multinomial probability judgment in geopolitical forecasting. An example of a multinomial prediction might be the probability that one of four candidates running for the same office wins the election. This prediction is represented by a probability distribution, with a simplex vector of probabilities of length four representing the probability each individual candidate is estimated to win, which cannot be collapsed into a singular value as in the binary case. To address this problem, a generalization of *WOA* was defined based on Euclidean distances between two probability distributions. Where $\mathbf{p}$ and $\mathbf{q}$ represent different simplex vectors of probability judgments across C possible outcomes, Euclidean distance is defined

$$D(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{c=1}^c (p_c - q_c)^2} \quad (4)$$

WOA can then be generalized into a distance-based metric, $DWOA$, which is given

$$DWOA = \frac{D(\mathbf{J}_1, \mathbf{J}_2)}{D(\mathbf{J}_1, \mathbf{J}_a)} \quad (5)$$

In the binary cases, $DWOA = aWOA$.

Himmelstein and Budescu (2023) found two additional egocentric effects, which may be specific to multinomial probability judgment. First, judges tended to revise their beliefs more frequently when the advice contained *counterfactual* information, where $\mathbf{J}_a$ ranked the likelihood of the discrete outcomes in a different order than $\mathbf{J}_1$. For example, judges were more likely to revise their beliefs when they initially thought Candidate A was the most likely winner, but the advice said Candidate C was more likely to win than Candidate A, and less likely to revise if the advice agreed that Candidate A had the best chance. Second, people revised more when the advice distribution was more *peaked* than their beliefs—that is, the advice put a larger probability magnitude on the most likely option than they did, which can be interpreted as a representation of advisor confidence. Measures of advisor confidence have also previously been shown to predict belief revision, but as these tasks did not involve multinomial probability judgment, advisor confidence was operationalized differently (Goodwin et al., 2013; Sniezek & van Swol, 2001; van Swol & Sniezek, 2005).

Continuous Probability Distributions

Most JAS research focuses on point estimates of people's beliefs. However, various measures of a judge's confidence have also been proposed to capture subjective uncertainty

(Sniezek & van Swol, 2001; van Swol & Sniezek, 2005). Changes in confidence are arguably just as relevant to defining belief revision as changes to point estimates. Recently, Soll et al. (2022) proposed a new measure called *influence of advice (IOA)* as a single metric that captures both revisions to point estimates as well as confidence following advice. *IOA* has two components, opinion shift and confidence change. Where μ_j is the mean of a probability distribution representing a judge's belief, and σ_j is its standard deviation, representing their confidence in that belief

$$Opinion\ Shift = \frac{(\mu_{J_1} - J_a)^2 - (\mu_{J_2} - J_a)^2}{(\mu_{J_1} - J_a)^2 + \sigma_{J_1}^2} \quad (6)$$

$$Confidence\ Change = \frac{\sigma_{J_1}^2 - \sigma_{J_2}^2}{(\mu_{J_1} - J_a)^2 + \sigma_{J_1}^2} \quad (7)$$

$$IOA = Opinion\ Shift + Confidence\ Change \quad (8)$$

One clear benefit of this approach is that it can capture *overconfidence*, which is another type of non-normative egocentric bias common in probability judgment (Aspinall, 2010; Colson & Cooke, 2018). In the JAS context, this can especially help measure the influence of confirmatory advice on overconfidence (Soll et al., 2022). However, the elicitation also requires a way to access not only a judge's point estimate, but an estimate of their uncertainty as well. Soll et al. (2022) propose several methods for doing so, including discretizing continuous values into C mutually exclusive bins, each of which is then assigned a probability (this can be considered a variation on multinomial probability judgment where the categories are ordered, see Himmelstein et al., 2021; Himmelstein & Budescu, 2023) as well as eliciting confidence intervals at certain percentile values (e.g., 95% confidence intervals). Given the added complexity of eliciting and analyzing judges' full subjective probability distributions, the present

research will continue to focus on point estimates and *WOA*, but future extensions including estimates of subjective uncertainty may be possible as well.

Continuous and Discrete Belief Revision

A common, but under-studied regularity in JAS research is that *WOA* often has a multi-modal empirical distribution. For example, Himmelstein and Budescu (2023) note that in each of their studies, participants only revised their beliefs around half the time. In the other half, $J_2 = J_1$, meaning they simply retained their initial belief. Though not often explored in detail, such patterns are common in publicly available JAS datasets (Logg et al., 2019; Schultze et al., 2015). Soll and Larrick (2009) identify a tri-modal pattern in belief revision, in which *WOA* has an extreme mode at 0, a wide distribution peaked somewhere between 0 and 1, and then a lower mode at 1, indicating $J_2 = J_a$, or cases where judges fully adopted advice².

The Dual Hurdle Model

Many studies attempt to understand advice utilization by modeling *WOA* using standard linear models (Bailey et al., 2023; Himmelstein & Budescu, 2023; Logg et al., 2019; Schultze et al., 2015). Himmelstein (2022) argued that, while this approach can help test certain broad hypotheses, the common trimodal distribution of *WOA* (Soll & Larrick, 2009) means several assumptions of such models are violated. In addition to being normatively limited to the interval $\in [0, 1]$, the tri-modality of *WOA* represents inflated values on the boundaries of 0 and 1. Methodologically, these boundary inflations can be accounted for by using a dual hurdle model, which posits that the decision process happens in two stages: first a judge decides whether to decline the advice entirely, adopt it in lieu of their prior belief, or seek a compromise between the two. If they choose to compromise, in the second stage they decide how much to weigh the

² These inflations at 0 and 1 are apparent whether *WOA* is Winsorized or not.

advice against their initial beliefs. Computationally, the model is a two-stage mixture model, with a multinomial logistic regression representing the initial discrete decision, and continuous regression representing the continuous second stage, conditional on compromise being selected in the first stage. If one relies on the normative assumption that constrains $WOA \in [0, 1]$ (Larrick & Soll, 2006; Soll & Larrick, 2009), Beta regression is an appropriate candidate for the second stage. The Beta distribution is traditionally parameterized with two shape parameters, α and β , but can be equivalently parameterized in terms of its expectation μ and precision ϕ as $\alpha = \mu\phi$ and $\beta = (1 - \mu)\phi$. Himmelstein (2022) relied on this normative assumption in applying the dual hurdle model to advice utilization, though conceptually the continuous mixture component could be based on any continuous distributional regression method.

Formally, the first stage of the model, for the initial discrete decision, can be written

$$WOA \begin{cases} = 0 \text{ with } P_{\text{decline}} = \eta_1 \\ = 1 \text{ with } P_{\text{adopt}} = \eta_2 \\ = c \text{ with } P_{\text{compromise}} = 1 - \eta_1 - \eta_2 \end{cases} \quad (9)$$

This stage of the model identifies the probabilities that participants make one of three discrete decisions: decline (P_{decline}), adopt (P_{adopt}), or compromise ($P_{\text{compromise}}$), with the probability space for the set of decisions defined by η_1 and η_2 . The second stage represents the case of compromise, given by c , or the degree to which the two sources of information are averaged³.

$$c \sim \text{Beta}(\mu\phi, (1 - \mu)\phi) \quad (10)$$

Because this model assumes $WOA \in [0, 1]$, it also constrains $0 < c < 1$. Each parameter can be conditioned on a linear combination of predictors (where $x = 1$ in the case of intercepts) as

³ Following the convention of Soll & Larrick (2009), in the context of advice utilization, the term “averaging” here and throughout this dissertation does not necessarily refer to simple unweighted averaging, but any possible weighted average (including simple averaging, or uniform weighting)

$$\eta_1 = \frac{e^{\beta_1 x}}{1 + e^{\beta_1 x} + e^{\beta_2 x}} \quad (11)$$

$$\eta_2 = \frac{e^{\beta_2 x}}{1 + e^{\beta_1 x} + e^{\beta_2 x}} \quad (12)$$

$$\mu = \frac{1}{1 + e^{-\beta_3 x}} \quad (13)$$

$$\phi = e^{\beta_4 x} \quad (14)$$

In cases where the judges make repeated measurements, the model can be defined hierarchically with random effects for each judge and/or item. The expectation of the model is given

$$E(WOA) = 0(P_{decline}) + 1(P_{adopt}) + \mu(P_{compromise}) = P_{adopt} + \mu(P_{compromise}) \quad (15)$$

Himmelstein (2022) re-analyzed several published JAS datasets using the dual hurdle model. One clear benefit of the model was that it was able to reveal that certain patterns which appear one way at the global level may actually be a result of a combination of different effects happening at different stages of the decision process. For example, the curvilinear effect of similarity between J_1 and J_a (Schultze et al., 2015) seemed to, in fact, be a product of two different effects. The proportion of cases where judges revised their beliefs (did not decline) was smallest when J_1 and J_a were most similar and increased monotonically as they became more dissimilar. However, the magnitude of belief revisions in cases where judges compromised showed the opposite effect: it was largest when J_1 and J_a were most similar, and decreased monotonically as they became more dissimilar (egocentric discounting). Thus, the curvilinear pattern in the aggregate appears to be a result of two separate processes: lower probability of belief revision for advice that was similar to people's initial beliefs, and egocentric discounting in belief revision magnitude for advice that was dissimilar from their beliefs. See Figures 5 and 6 of Himmelstein (2022) for an illustration.

One important implication of the boundary inflation at $WOA = 0$ is it requires the initial belief as a reference point, not just computationally but psychologically. In cases where people decline advice, how would they have behaved if they hadn't formalized their beliefs in advance? Would their beliefs after advice exposure match what their naïve pre-advice beliefs would have been, or does the act of explicating one's belief alter the process entirely?

Truncated JAS Studies

While the counterfactual question raised about belief *change* in the absence of initial measurement has never been addressed, the question of how posterior beliefs reflect advice has received some attention in the literature. As part of a more general study about the effects of different sources of information cueing on belief, Sniezek et al. (1995) studied this problem in a series of binary choice tasks. An example item involved asking judges to correctly estimate whether “41% of all money spent in the US is spent in (A) restaurants or (B) supermarkets” (p 165). Participants in one condition followed the standard JAS procedure: they gave an initial answer (J_1), saw the estimate of an independent advisor (J_a), then made a second judgment (J_2). For participants in another condition, the initial judgment was truncated. They simply saw the advice, then provided their response, without ever making an initial guess. Results indicated judges tended to be more accurate in the untruncated (standard JAS) condition than the truncated condition.

Yaniv and Chosen-Hillel (2012) conducted a study in which subjects in a “blindfold” condition made prior predictions but did not have the content of the target disclosed until after viewing advice. For example, Study 1 featured a “full view” JAS condition, where participants were asked to estimate the number of calories in an orange, then given advice, then allowed to revise. However, in the blindfold version of the item, subjects were asked to estimate the number

of calories in “Food K,” then given advice, then allowed to revise. In Study 2, both the full view and blindfold condition had the prior belief estimates truncated, as in Sniezek et al. (1995). In Study 3, the JAS was also truncated, but judges in the blindfold condition were allowed to revise after having the true target revealed. While this study did not directly compare the traditional JAS paradigm to a truncated version, the purpose was similar: to determine how advice would be taken in the absence of a prior anchor that was encoded based on one’s knowledge about the judgment target. Their results indicated that participants in the blindfold condition tended to make posterior judgments that were more similar to advice, and thus produce more accurate posterior estimates.

Koehler and Beauregard (2006) conducted three studies primarily focused on comparing how similar people rated their own judgments to that of an advisor depending on whether advice exposure happened before (exposed condition) or after (unexposed condition) they made their own judgments. Their studies involved estimating the years during which notable historical events occurred, such as the fall of the Roman empire, or the beginning of the Ming dynasty in China. They found that when advice exposure preceded judgment, participants rated the advice as more similar to their own independent beliefs than when their judgment preceded advice exposure. In their final study, they compared a variant of the truncated JAS to a more typical untruncated JAS and found posterior belief tended to be more similar to advice in the truncated than untruncated condition.

Rader et al. (2015) provided the most direct and comprehensive attempt to date to answer the question of how beliefs reflect advice in a truncated vs untruncated JAS. In the standard, untruncated version, participants were asked to guess the age of a person in a photograph (J_1).

They then saw an advisor's guess (J_a) and were allowed to revise their estimate (J_2)⁴. In the truncated version, J_1 was never elicited. Instead of studying accuracy, as Snizek et al. (1995) did; or mean similarity to advice, as Koehler and Beauregard (2006) did; Rader et al. (2015) examined the full shape of the distribution of J_2 across the two conditions. They found a clear and robust pattern across participants: the distribution of J_2 tended to move away from J_a in the truncated condition, while it tended to peak at J_a in the untruncated condition. Rader et al. (2015) concluded that when the JAS was truncated, people used J_a as a reference point and attempted to falsify it, a pattern that does not occur in standard (untruncated) JAS studies. Interestingly, Rader et al. (2015) found the accuracy of J_2 was similar in both conditions, on average, despite the clear differences in the distributions.

These results conclusively demonstrate that elicitation of a prior belief changes the psychological process of advice utilization. However, because J_1 was never elicited in the truncated condition, their method cannot provide a full psychological account of advice utilization, nor can it measure the magnitude of the belief change. It can help us make inferences that use J_a as a reference point, but fully understanding such change requires both J_1 and J_a as reference points.

Planned Missing Data

One of the hottest topics in psychometrics involves the handling of missing data (Little & Rubin, 2019; van Buuren, 2018). Most approaches are aimed at cases where the data may be missing incidentally, such as when participants skip questions, study length leads to attrition or dropout, or participants are unable or unwilling to answer certain items. The literature describes

⁴ This specific paradigm, involving guessing the age or weight of a person in a photogram, is common in JAS studies (e.g., Logg et al., 2019; Schultze et al., 2018).

three mechanisms by which data may be missing: missing completely at random (MCAR), missing at random (MAR), or not missing at random (NMAR). MCAR refers to scenarios when there is no causal mechanism leading to missing data—it is unconditionally random. In such scenarios, the missing data do not bias the results, and methods for handling them are typically aimed at improving statistical power. MAR refers to cases where there is a systematic reason some data are missing, but that reason can at least partially be modeled using other variables available in the dataset. It is necessary to account for these confounds to avoid bias when data is MAR, but it is possible to do. NMAR refers to cases where data are systematically missing, but in a way that cannot be modeled based on other variables in the dataset and, as a result, bias becomes unavoidable.

Researchers have also successfully used *planned missing data* studies to reduce study costs and participant burden without sacrificing statistical power (Graham et al., 2006; Rhemtulla & Hancock, 2016; Silvia et al., 2014). In planned missing data designs, researchers require responses to a large set of items that may be too long or expensive to administer. Doing so could also lead to participant fatigue, attrition, and poor data quality. Instead, the item bank is broken up into shorter, overlapping forms that are randomly administered to different groups of participants. Information about how items are correlated allows researchers to treat items omitted from a given form as missing data and impute estimates of what subjects' responses to the missing items might have been. Because the omitted items are selected randomly, they are, by definition, MCAR equivalent (there is no causal mechanism for missingness beyond random assignment) and can be imputed without introducing bias. There are several statistical methods for modeling missing data, including maximum likelihood (Enders, 2001), multiple imputation (Sinharay et al., 2001), and fully Bayesian methods (Gelman et al., 2013; Ma & Chen, 2018).